

Week 4

Keel Bone Chicken Project

02/06/26



College of Engineering

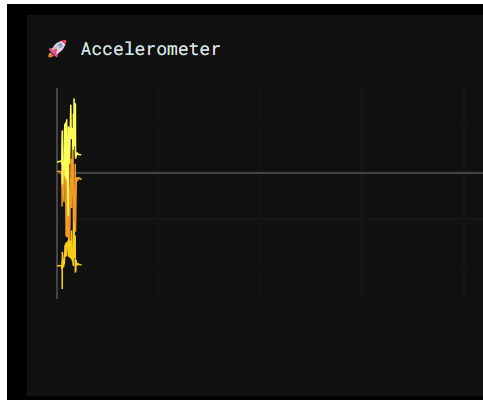
This Week

Weekly Progress

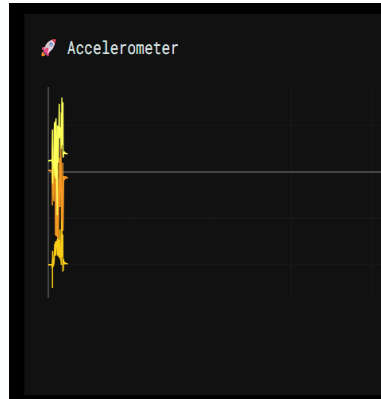
- **Devices Prepared:**
 - Devices all successfully flashed
 - Ready hand over to Agriculture department (Wednesday)
- **Website Updated:**
 - Targeting time changed to 24 hours; live website changed to match
- **Sample Data Collected:**
 - IMU Alarm data collected
 - Analysis of data
- **Purchase Reports:**
 - Sensor form 12(s)
- **Transmitter Circuit**
 - Creation of circuit and PCB

Devices Prepared

Devices all successfully flashed at varying rates



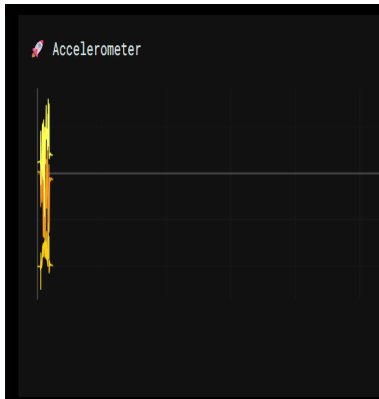
10 Hz



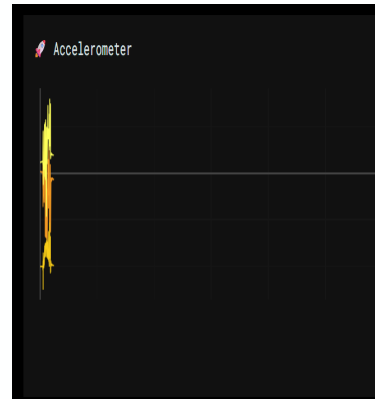
25 Hz



50 Hz



100 Hz

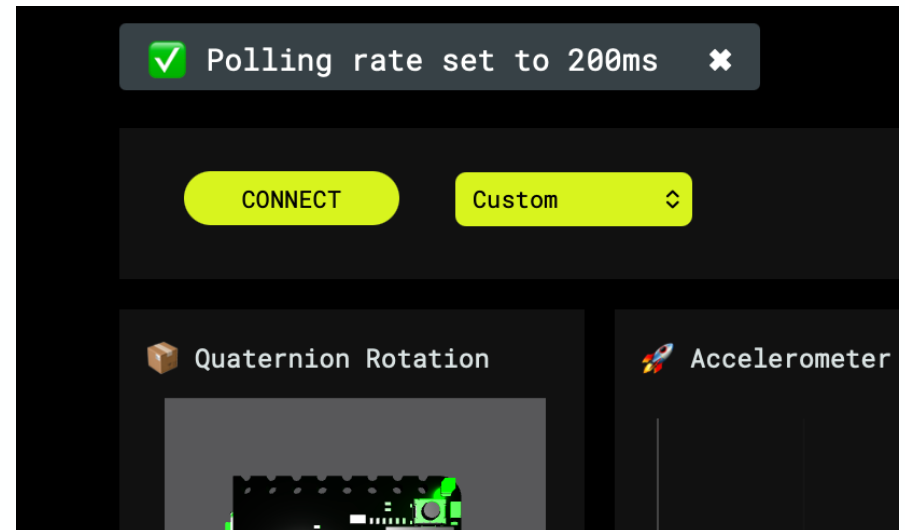
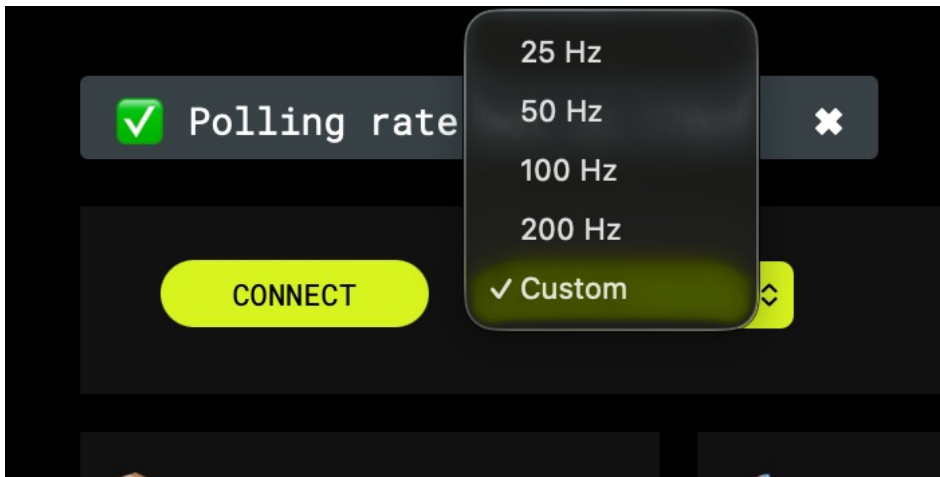


200 Hz

Devices Prepared

Ready hand over to Agriculture department (Wednesday)

- Sampling Rate can now be adjusted before collecting data
 - In addition to changing sampling rate in firmware
 - Adds additional layer of verification
- Easy and interactive for Agriculture department to use:



Website Updated

Targeting time changed to 24 hours; Live website changed to match

- Verified through continuous 20-24 hour data collection period
- No apparent adverse effects
- Longer image spread:



Sample Data Collected

IMU Alarm Data Collected

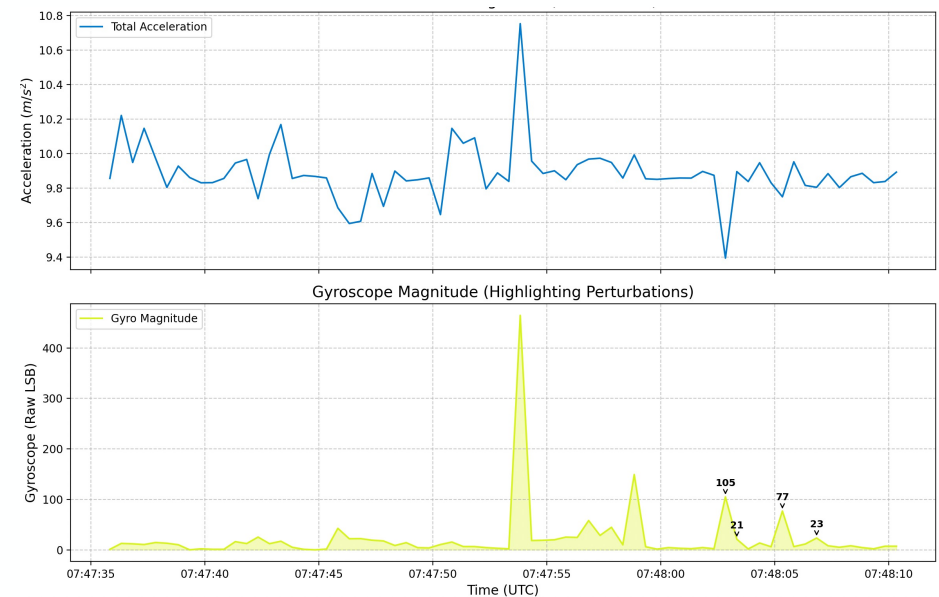
- Slight perturbations from IMU visible from alarm vibrations
- Test run and averaged over all 5 sampling rates



Sample Data Collected

Analysis of data

- Python script to analyze if the small perturbations are correct
- Seems initial vibration causes a major initial spike
 - 9.8 -> 10.8 m/s²
- Conversion rate: 1g ~ 4096 LSB
 - Bit notation of the senso gets converted to raw acceleration



Sample Data Collected

Program used for data capture

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

def analyze_nicla_data(file_path):
    # 1. Load the data
    try:
        df = pd.read_csv(file_path)
    except FileNotFoundError:
        print(f"Error: File '{file_path}' not found.")
        return

    df['timestamp'] = pd.to_datetime(df['timestamp'])

    # Sort data chronologically (CSV is often reverse chronological)
    df = df.sort_values('timestamp').reset_index(drop=True)

    # 3. Unit Conversion
    # Accelerometer: 1g is approx 4096 LSB (based on Z-axis ~4100 at rest)
    # 1g = 9.80665 m/s^2
    lsb_to_g = 1 / 4096.0
    g_to_ms2 = 9.80665

    # Calculate components in m/s^2
    df['Ax_ms2'] = df['accelerometer_Ax'] * lsb_to_g * g_to_ms2
    df['Ay_ms2'] = df['accelerometer_Ay'] * lsb_to_g * g_to_ms2
    df['Az_ms2'] = df['accelerometer_Az'] * lsb_to_g * g_to_ms2
```


Sensor form 12(s)

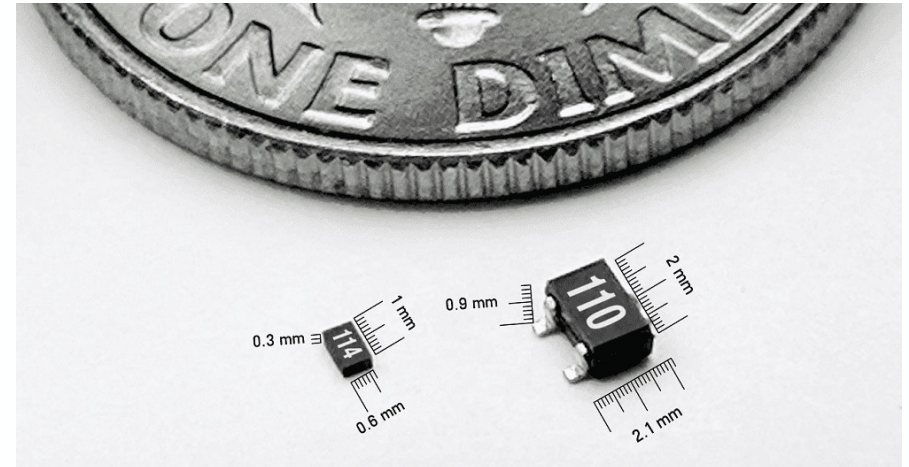
12/08/25		DATE PRICE OBTAINED:	QUOTE VALID END DATE:		
ACCOUNT(S) TO BE CHARGED					
PRF ACCOUNT	ORDER/WBSE			G/L ACCOUNT	AMOUNT
ON	PART NUMBER	QUANTITY	UNIT PRICE	EXTENDED PRICE	Total Expenses
nsor	Y3-9556688A	2	40	100	100
					0
					0
					0
					0
					0
					0
					0
					0
					0
					0
					0
					0
					0
					0
				Total:	100
CP:			PURCHASING CENTER USE ONLY X		

- Form 12 for force sensor

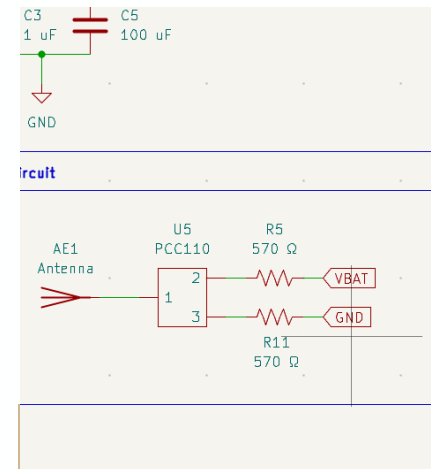
Transmitter Circuits

KiCAD Transmitter PCB and Circuits

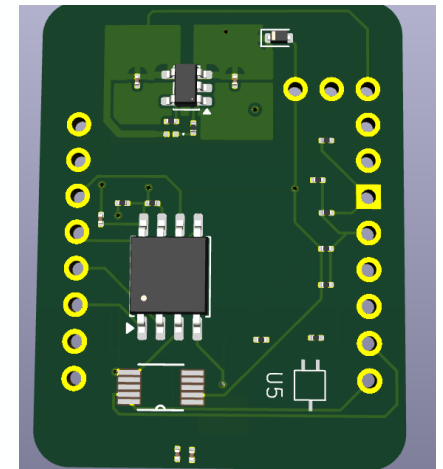
- Created custom footprint for PCC110
 - Changed footprints and circuit
 - Updated PCB location and wiring
- Found suitable Antenna for PCC110 Harvester
- Verified with Powercast support



PCC110 Sizing



PCC110 RFID Circuit



PCC110 RFID PCB

Next Week

Plans & Blockers

- More Data Analysis of IMU
- Obtain Powercast RFID -> DC Converters from Agriculture
- Verify Powercast RFID circuits on breadboard
- Research Papers:
 - IEEE
 - Transactions on Electronics Circuits and Systems
 - 2 – 5 pages
 - 1 – Long Journal
 - IoT Journal <-
 - Transactions on Industrial Electronics
 - Electronics DPI

Thank You

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